

A UAV-Assisted Truth Discovery Approach With Incentive Mechanism Design in Mobile Crowd Sensing

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Abstract—Incentive mechanisms are essential to incentive workers carrying mobile handheld devices to participate in mobile crowd sensing and finally achieve good truth discovery performance. However, malicious workers may report false or malicious data to defraud rewards, resulting in service quality degradation. Moreover, the existing incentive mechanism is challenging to identify malicious workers when recruiting workers in reality, which results in low accuracy of truth discovery and waste of cost. In this paper, we propose an Incentive-based Truth Discovery (ITD) scheme to incentive credible workers to submit high-quality data, thereby enhancing the accuracy of truth discovery. In the ITD, an unmanned aerial vehicle (UAV)-assisted split-aggregation truth discovery mechanism is proposed firstly to infer the truth. The addition of the UAV can improve the accuracy of truth discovery and assist in evaluating workers' trust. Then, we evaluate the quality of participants' data and propose a data quality-based trust meter to update each worker's trust to guide future recruitment efforts. Finally, a Quality-aware Trustworthy Incentive (QTI) mechanism is proposed to select credible workers for data collection and provide them with reasonable payments. The experimental results show that ITD improves the accuracy of truth discovery by 98.47%, over the state-of-the-art, at a sensing cost reduced by as high as 43.89%.

Index Terms—Mobile crowd sensing, truth discovery, incentive mechanism.

I. INTRODUCTION

MOBILE Crowd Sensing (MCS) [1], [2], [3] is an emerging application paradigm that utilizes mobile devices with rich perception capabilities, such as smartphones and smartwatches, for data collection to discover good truths.

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MCS comprises two essential components: the platform and workers. The platform first releases the time, location, and data types of tasks. Subsequently, workers who carry devices with many sensors, such as cameras, microphones, accelerometers, and GPS, voluntarily participate in data collection to complete these tasks. This collective effort generates vast data, enabling the platform to aggregate the data to discover good truths that can support applications requiring sensing in an extensive area or over a long period with low costs [4], [5], [6]. So far, various MCS applications have been used in different fields, such as NoiseTube [7] for noise monitoring, CrowdAtlas [8] for traffic monitoring, and Parknet [9] for finding on-street parking spots.

Discovering the accurate estimated truth is the basis for the platform to make accurate decisions and then provide workers with thoughtful services [10], [11], [12]. However, the presence of malicious workers can affect the results of truth discovery. Because during the data collection process, workers need to move to a designated location to collect data, which consumes workers' time and mobile devices' system resources (such as power, computing power, and network bandwidth) [13]. That makes it possible for malicious workers to report low-quality or even false data to gain extra income [14]. When the data received by the platform contains false or low-quality data, estimations would stray away and incur non-negligible errors [15]. Especially, workers are distributed in different geographic locations, making it expensive for the platform to verify the authenticity of the collected data. In some cases, the authenticity cannot be verified. For example, workers distributed across several thousand kilometers will find migratory birds and obtain data such as images and videos when monitoring migratory birds [16]. However, migratory birds may stay in a particular place for only a few minutes, and data on migratory birds cannot be obtained after a few minutes, which leads to the data collected by workers not being verified. Unverifiability of data may provide opportunities for malicious workers to submit false or malicious data, further exacerbating their influence on truth discovery. Therefore, there is an urgent need for a method to verify the data provided by workers and identify malicious workers, thereby enhancing the accuracy of truth discovery.

To get the accurate estimated truth, some researchers have proposed recruiting enough workers to collect data for the same task. They apply the average method [17], [18], the median method [18], voting [19] or the weighted average

method through the Conflict Resolution on Heterogeneous [20], [21], [22] to obtain the estimation. However, recruiting more workers will produce higher sensing costs since the crowd-sensing campaigns involve monetary costs [23], [24], [25]. Moreover, the estimated truths obtained by these methods are easily influenced by malicious workers. Therefore, other researchers did not directly recruit workers but evaluated the workers' trust [26], [27]. They proposed many trust evaluation methods, including Bayesian-based trust evaluation [28] and cloud theory-based trust evaluation [29], etc. However, the existing methods obtain trust evidence by passively observing, thus having defects such as difficulty obtaining evidence, inaccurate results, and limited applications. Furthermore, to the best of our knowledge, these methods either assume the quality or authenticity of the data submitted is verifiable or assume that workers' trust has been obtained [30]. Nevertheless, the data authenticity and the workers' trust are challenging to acquire in reality.

In addition, the above truth discovery methods assume that the platform has collected enough data to explore the truth but ignore the impact of the incentive mechanism on discovering accurate truth. Incentivizing workers to submit high-quality data is critical for the platform to aggregate and extract information to improve the accuracy of the estimated truth [10], [11], [12]. In the existing incentive mechanism literature, some researchers paid for workers according to the quality of their data [10], [31]. However, the lack of benchmarks to measure the reliability of the sensing data and the presence of malicious workers lead to inaccurate assessments of data quality reported by workers while generating unnecessary cost overhead. Therefore, it becomes imperative to contemplate an incentive mechanism for truth discovery aimed at recruiting and incentivizing credible workers to contribute high-quality data, thereby discovering the accurate truth facilitating.

Motivated by the above discussion, in this paper, we aim to proactively evaluate workers' trust through a realistic and feasible scheme and incentive workers to submit high-quality data to discover more accurate truths. Along with the increasing popularity of the Unmanned Aerial Vehicle (UAV), it is foreseen that they will play a key role in MCS. With its rapid deployment and controllable mobility capabilities, UAVs can autonomously crowd sensing anytime, anywhere. In particular, UAVs equipped with rich sensors could act as data providers to sense regions that could hardly be reached by terrestrial vehicles, e.g., flood hazard areas or nuclear-polluted areas. However, just using the UAV to collect data to obtain accurate truths will be extremely costly [32], [33]. Despite this, it provides inspiration for verifying data accuracy and proactively assessing workers' trust.

Therefore, we propose an Incentive-based Truth Discovery (ITD) scheme in this work. As we all know, the data submitted by trusted workers would be close to the ground truth and present higher quality. In ITD, we verify the accuracy of the data and evaluate workers' trust actively by using the UAV assisted. In this process, considering the cost of the UAV, we select trusted workers as proxy workers to evaluate the trust of others. Then, choose to recruit credible workers and incentivize them to submit high-quality data to the platform.

Finally, the platform infers the truth of the task based on the collected data and existing knowledge. In summary, the main contributions of this work are as follows:

- To obtain a more accurate truth about the task, we propose a UAV-assisted split-aggregation truth discovery mechanism. Specifically, the workers recruited should be classified according to the worker's trust first, and then we obtain the estimation by minimizing deviation from the truths to the multi-proxy worker input.
- We evaluate the data quality of participating workers by the data collected from the UAV and propose a data quality-based trust meter to update the trust of each worker. It guides the recruitment of the next round of workers and provides a reliable basis for classifying workers.
- The Quality-aware Trustworthy Incentive (QTI) mechanism is designed to select trusted workers to collect data in the long run and assign rewards to workers according to their data quality, which effectively improves the quality of MCS applications. The experimental results show that the ITD scheme proposed in this paper improves the truth discovery accuracy by 98.47% compared with the Conflict Resolution on the Heterogeneous (CRH), and the cost of the proposed QTI is reduced by 43.89% compared with the CRH.

The rest of this paper is organized as follows. The related work is reviewed in Section II. The system model and problem statement are introduced in Section III. In Section IV, we introduce the design details of ITD, and Section V provides the experimental results. Section VI concludes this paper.

II. RELATED WORK

A. Truth Discovery

Truth discovery is used to effectively discover the accurate truth from noisy or even conflicting sensory data. Some researchers assume that the platform already has enough data and mainly addresses effectively discovering the truth. Simple aggregation strategies to obtain the truth through traditional methods (such as simple average, median, or voting) [17], [18], [34] treat all workers equally and may not provide reliable aggregation results. Because in the MCS system, the information in sensory data provided by workers usually varies greatly among different workers. Although there are studies [32] using UAV swarms to collect data, relying solely on UAV swarms to collect data will incur high costs, which is not sustainable. Thus, some works proposed the iterative method to discover the truth [20], [21], [22], [35], [36], [37], [38], [39]. Compared with traditional methods, the accuracy of truth discovery has been significantly improved.

Among them, TruthFinder [37] is proposed to ensure the correctness of the information on the internet and consider that different websites often provide contradictory information on a topic, such as different specifications for the same product. The authors exploit the relationship between websites and information, using the iterative method to infer the website's credibility and the information's correctness between them. Accu [35] is proposed to solve how to discover the truth from conflicting information that exists from a large number of

sources, some of which may be copied from other sources. The authors use Bayesian analysis to identify dependencies among workers and design an algorithm that iteratively detects dependencies to discover the truth from conflicting information. Galland et al. [36] proposed three algorithms, Cosine, 2-Estimates, and 3-Estimates, to estimate the truth of facts and trust in workers. Wang et al. [39] proposed a method based on maximum likelihood estimation. The method estimates the reliability of each participant and the truthfulness of each measurement through an expectation-maximization algorithm until convergence. CRH [20], [21], [38] is proposed to obtain a more accurate truth. Specifically, they iteratively estimate the truth using the weighted aggregation method, updating workers' weights (reliability) and the estimation until convergence. The estimate of the truth is naturally close to the data observed by workers with higher weights. However, these truth discovery methods are vulnerable influenced to malicious workers, and whether workers provided positive statements did not guide future tasks. Moreover, they did not take into account the impact of incentives on truth discovery.

B. Incentive Mechanism

To collect data, workers need to move to designated locations to perform sensing tasks, which consumes power, communication resources, and time on workers' devices [24]. Incentive mechanisms provide the basis for more accurate truth discovery. Therefore, an appropriate incentive mechanism needs to be designed to incentive workers to participate in tasks. We divide existing incentive mechanisms into two categories. One class mainly incentives enough workers to participate in tasks, which we call quantity incentives. Another class of incentives for workers to submit high-quality data for tasks, we call quality incentives. However, these incentive mechanisms did not mention how the corresponding truth discovery mechanism should be designed.

In order to incentive enough workers to participate in the task, many studies on incentive mechanisms [23], [40], [41], [42], [43], [44], [45]. Most of them use the auction or game theory models to deal with the strategic behavior of workers, and the design goals of these incentive mechanisms are different. Specifically, Baek et al. [23] proposed an expected social welfare maximization mechanism, which aims to attract and retain more participants and ultimately achieve higher social welfare and platform utility. Zhang et al. [41] proposed the MobiAuc mechanism, which aims to provide a mechanism to incentive workers to participate and ensure fair transactions. Wang et al. [40] proposed to incentive smartphone workers to join a mobile crowdsourcing application taking into account workers' location information. Hussein et al. [44] introduces a new service called IVWaaS that utilizes moving vehicles on the road as witnesses to specified events. Mainly, they propose privacy-conscious incentives to facilitate the widespread adoption of the service. However, the more workers the platform recruits, the higher the cost is.

Therefore, another class of research mainly incentivizes workers to submit high-quality data [10], [31], [46]. Specifically, Dan et al. [10] use Maximum Likelihood Estimation

(MLE) to indicate the quality of perceived data through convergent estimation of the participant effort matrix. Wen et al. [31] proposed a Quality Driven Auction, where the worker is paid based on the quality of sensed data to incentive them to submit high-quality data. Jin et al. [46] proposed a new framework, Thanos. It is a reverse auction-based incentive framework for quality perception to incentive worker participation in the MCS, and it considers the quality of sensing data contributed by individual workers to ensure near-optimal social welfare. However, these studies are based on the assumption that the authenticity of the data can be verified. In fact, the authenticity of the data is difficult to verify, and the lack of benchmarks to make the final evaluation of the quality of the worker's data inaccurate, resulting in unnecessary cost overhead.

C. Incentive Mechanism for Truth Discovery

Some existing studies focus on the mechanism of truth discovery and consider the influence of incentive mechanisms on truth discovery simultaneously to improve the accuracy of truth discovery further [22], [25], [47]. Specifically, Zhang et al. [25] proposed to incentivize crowd workers to undertake binary labeling tasks and used a label aggregation technique based on a sequential Bayesian approach to determine the true answer, considering that workers of different levels may provide labeled answers with different reliability and the task difficulty may differ. Jiang et al. [47] designed incentive to discover the truth about text answers through the photocopier. They propose a truth discovery algorithm based on Bayesian analysis, which calculates the dependencies between each pair of workers and estimates the authenticity of each task based on the worker's dependencies and accuracy. Wang et al. [22] designed an incentive mechanism under discretely personalized privacy awareness considering the worker's privacy levels, and they mainly use CRH [20] to obtain good truth discovery performance. However, they did not take into account the impact of malicious workers on truth discovery, where the presence of malicious workers can lead to inaccurate assessments of data quality reported by workers while generating unnecessary cost overhead and resulting in low accuracy of truth discovery.

For convenience, we listed several main methods and compared whether these methods consider truth discovery, malicious workers, and incentives in TABLE I. Obviously, our method not only considers the existence of malicious workers but also combines truth discovery with incentives to discover more accurate truth.

III. SYSTEM MODEL AND PROBLEM STATEMENT

A. System Model

In this paper, we consider a typical MCS system, which consists of two basic components: the platform and the workers. The main goal of the platform is to publish sensing tasks, aggregate data from workers to obtain accurate truth, and reward them based on reported sensing data. Workers are distributed around the city and equipped with multiple smart data reporting devices such as sensors, surveillance

TABLE I
MAIN METHODS COMPARISON

Methods	Truth Discovery	Malicious workers	Incentives
TruthFinder [37]	Yes	No	No
Accu [35]	Yes	No	No
3-Estimates [36]	Yes	No	No
CRH [20]	Yes	No	No
QM [10]	No	No	Yes
QDA [31]	No	No	Yes
[22]	Yes	No	Yes
DATE [47]	Yes	No	Yes
[25]	Yes	No	Yes
Ours	Yes	Yes	Yes

probes, mobile phones, laptops, etc. These mobile devices have powerful communication and storage capabilities. As data collectors, workers voluntarily participate in data collection and then upload the data to the platform to obtain rewards. During the above process, malicious data collectors may submit false or malicious data, resulting in low accuracy of truth discovery. Therefore, we propose the ITD that can accurately distinguish normal workers from malicious attackers through UAV-assisted active trust detection and incentive workers to submit high-quality data to discover accurate truth. Assume that there are M workers, which is represented as $\mathcal{M} = \{1, 2, \dots, M\}$. They aim to complete N tasks published by the platform, represented as $\mathcal{N} = \{1, 2, \dots, N\}$.

B. Attack Model

In the attack model, we considered mainly malicious workers reporting false data or low-quality data in MCS. In this paper, we mainly consider three typical attacks, namely probabilistic, on-off, and data tampering attacks. Probabilistic attacks refer to malicious workers submitting low-quality data probabilistically. Data tampering refers to malicious workers tampering with data, forging false data so that there is a big difference between the data submitted to the platform and the ground truth. The on-off attack is when the malicious worker alternates between bad and good behavior. In addition, extending the method to other attacks has not yet been considered due to the variety of malicious attacks. In response to these attacks and to obtain accurate truth discovery, we introduced ITD.

C. The Framework of ITD

The framework of ITD is shown in Fig. 1, where we use the reverse auction framework in the incentive mechanism to describe the incentives of the platform for workers in ITD. Firstly, the platform publishes sensing tasks to all idle workers, and then these idle workers bid for the task they are interested in. After bidding, the platform selects a certain number of highly trusted workers to complete the task, after which the selected workers collect data and submit the data to the platform. Truth discovery occurs to obtain the estimation of truth for the task after the platform has collected sensory data submitted by all participating workers. Finally, the platform evaluates the data quality of workers and then updates their trust to guide the next round of worker recruitment. Ultimately,

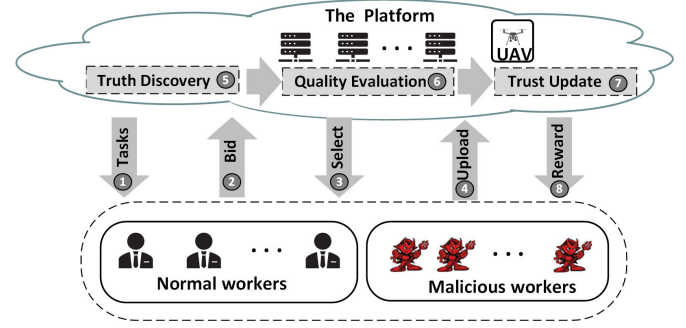


Fig. 1. The framework of ITD.

TABLE II
PARAMETERS IN SYSTEM MODEL

Parameter	Meaning
$\mathcal{N}$	The set of tasks
$\mathcal{M}$	The set of workers
$\bar{\omega}_{nor}$	The average trust of normal workers
$\bar{\omega}_{mal}$	The average trust of malicious workers
C	Total system cost
x_n^*	The estimation of the n -th task
x_m^n	The data of the m -th worker about the n -th task
ω_m	The trust of the m -th worker
$q_{m,n}$	The data quality of the m -th worker about the n -th task
h_m	The weight of the m -th worker
φ	Attenuation factor
Θ	Threshold of the number of tasks collected by the UAV
c_u	The cost of the UAV acquisition per unit task
γ	Trust threshold
δ	Negative factor
μ	Positive factor
ε	Quality threshold
bid_m	The bid of the m -th worker

the platform pays all participating workers. It is worth mentioning that the addition of the UAV is mainly used to assess workers' data quality, and the specific operation is detailed in Section IV. The notation table is shown in TABLE II.

D. Problem Statement

The objective of the MCS system is to incentive workers to collect and submit high-quality data and ultimately extrapolate the ground truth of the task from the data collected by all workers. Previous research has shown that the accuracy of truth discovery is closely related to the data quality reported by workers. However, in practice, some workers may inadvertently submit low-quality data for equipment reasons, and some may even provide fake data to receive additional rewards. That leads the result of truth discovery to be inaccurate or even wrong. Therefore, evaluating workers' trust is the key to obtaining accurate results of truth discovery.

The first problem we tackle is effectively distinguishing normal workers from malicious ones. The MCS platform should update workers' trust based on the quality of data reported by the worker. The second problem is to maximize the accuracy of aggregation. The aggregation error is denoted as the distance between the estimated value and its ground truth. The smaller the aggregation error, the greater the aggregation accuracy, the closer the estimation is to the ground truth, and the more effective the truth discovery method is. The third

issue is to minimize the cost of truth discovery. The cost of the system mainly contains the costs of hiring workers for data collection and the cost of dispatching the UAV for data collection. The performance of the system should be evaluated through the following indicators:

(1) The difference of trust between normal and malicious workers D is shown in Eq. (1).

$$D = \overline{\omega_{nor}} - \overline{\omega_{mal}}. \quad (1)$$

Among them, $\overline{\omega_{nor}}$ and $\overline{\omega_{mal}}$ refer to the average trust value of normal workers and that of malicious workers, respectively. The difference in the trust values between the two types of workers shall reflect the effectiveness of the proposed trust evaluation mechanism. Therefore, one of the goals of this paper is $\max(D) = \max(\overline{\omega_{nor}} - \overline{\omega_{mal}})$.

(2) The mean absolute error (MAE) between the estimation and the ground truth on N sensing tasks:

$$MAE = \frac{1}{N} \sum_{n=1}^N |x_n^* - x_n^{truth}|. \quad (2)$$

where x_n^{truth} is the ground truth of the n -th sensing task, x_n^* is the estimation of the n -th sensing task. We use MAE to represent the aggregation accuracy. The smaller the MAE, the higher the aggregation accuracy. x_n^{truth} is just to evaluate the performance of the system. Therefore, the second goal is $\min(MAE) = \min(\frac{1}{N} \sum_{n=1}^N |x_n^* - x_n^{truth}|)$.

(3) The total cost is represented by:

$$C = \sum_{n=1}^N (g_{n,u} \cdot c_u + \sum_{m=1}^M g_{n,m} \cdot p_{n,m}). \quad (3)$$

The total cost mainly includes the cost of the UAV and hiring workers. Eq. (3) represents the cost of performing one round, where $g_{n,u} = 1$ or 0 indicates whether the UAV collects the data of the n -th task and c_u represents the cost of the UAV acquisition per unit task. Similarly, $g_{n,m} = 1$ or 0 indicates whether the m -th worker participates in the data collection of the n -th task. $p_{n,m}$ represents the incentive cost for the m -th worker to perform the n -th task. Hence the third goal of our strategy is: $\min(C) = \min(\sum_{n=1}^N (g_{n,u} \cdot c_u + \sum_{m=1}^M g_{n,m} \cdot p_{n,m}))$.

IV. DESIGN DETAILS OF ITD

In this section, we proposed a UAV-assisted split-aggregation truth discovery mechanism to obtain more accurate truth. Then, we quantified the data quality of workers, and a data quality-based trust meter was proposed to update the trust of workers based on this. Finally, we put forward the QTI mechanism to recruit credible workers and incentive them to submit high-quality data.

A. Truth Discovery

Truth discovery has recently become an effective tool in the MCS system since it can discover the truth of tasks from noisy and conflicting perceptual data. As is known to all, the higher the workers' trust, the more accurate the report data. Moreover, the estimated truth of a task can be affected in the

following two aspects. On the one hand, malicious workers will launch attacks to submit malicious data, which impacts the estimated truth. On the other hand, there will be differences in the data submitted by normal workers due to the influence of factors such as equipment quality. Considering the above problems, we propose a UAV-assisted split-aggregation truth discovery mechanism, which first classifies workers according to the execution of the task and then infers the truth. The specific process is described as follows.

As a data collection tool, the UAV is used in the assessment. When the UAV moves to a specific position, it accurately collects the sensing data within its communication range. Data collected by workers and the UAV will be sent to the data center, where it is stored, calculated, and processed. The sensing data collected by the UAV is considered authentic and credible, as it is less affected by environmental noise and dispatched by a trusted platform. The data collected by the UAV can be directly used as ground truth when the UAV collects task data, i.e., $x_n^* = V_n$. x_n^* and V_n are represent the estimated truth of the n -th task and the data about the n -th task collected by the UAV, respectively. Moreover, x_n^* can be used as baseline data to evaluate the quality of data collected by workers.

However, in data collection by the UAV, it is difficult to collect data for all tasks due to cost constraints, resulting in a limited amount of data provided by the UAV. Therefore, we use workers with higher trust than a threshold γ as proxy workers and set the number of tasks collected by the UAV not to exceed a given threshold Θ . In addition, our work proposes to split and aggregate based on trust to obtain a more accurate estimate of the truth. It performs the implementation of data acquisition staff classification during the data aggregation process. If there are proxy workers, the truth can be inferred based on the data collected by proxy workers. On the contrary, if there are no proxy workers, the truth can be inferred based on the data collected by all workers recruited. It is worth mentioning that the UAV will collect data on the task that is collected with no proxy workers in this paper. The specific process is as follows.

Suppose there are some proxy workers among the recruited workers to perform the task. In that case, we obtain the estimation by minimizing deviation from the truths to the multi-proxy worker input where the weight reflects the data contribution degree of workers. Specifically, our optimization objective is expressed as follows.

$$\min f(x_n^*, \mathcal{H}) = \sum_{m \in \mathcal{M}_n^p} h_m d(x_n^m, x_n^*), \quad (4)$$

$$s.t. \sum_{m \in \mathcal{M}_n^p} \exp(-h_m) = 1. \quad (5)$$

where x_n^m , $\mathcal{M}_n^p$, and $h_m(\mathcal{H} = \{h_m\}_{m \in \mathcal{M}_n^p})$ represent the data of the n -th task collected by the m -th worker, the set of all proxy workers executing the n -th task, and the weight of the m -th worker, respectively. $d(x_n^m, x_n^*)$ is the loss function, which represents the distance between x_n^m and x_n^* . When the data x_n^m deviates from the aggregated data x_n^* , the loss function value is relatively high; when x_n^m is close to x_n^* , the loss function value is relatively low. We apply Lagrange

multipliers to solve this optimization problem as follows.

$$L = \sum_{m \in \mathcal{M}_n^p} h_m d(x_n^m, x_n^*) + \lambda \left(\sum_{m \in \mathcal{M}_n^p} \exp(-h_m) - 1 \right), \quad (6)$$

where λ is a Lagrange multiplier. Let the partial derivative of with respect to is 0, and then we can obtain the solution of h_m as follows:

$$h_m = -\log \left(\frac{d(x_n^m, x_n^*)}{\sum_{m \in \mathcal{M}_n^p} d(x_n^m, x_n^*)} \right). \quad (7)$$

The specific truth discovery process is mainly divided into two parts: Weight Estimation and Truth Inference.

Weight Estimation: In this step, workers' weights are calculated based on the currently estimated truth and the data they reported. Specifically, the weight h_m of each worker is calculated by Eq. (7).

Truth Inference: This step conducts weighted aggregation based on the estimated worker weights for the sensing task.

$$x_n^* = \frac{\sum_{m \in \mathcal{M}_n^p} h_m x_n^m}{\sum_{m \in \mathcal{M}_n^p} h_m}. \quad (8)$$

The truth discovery process of proxy workers performing the n -th task is shown in Algorithm 1. Truth discovery starts with the initialization of workers' weights and then iteratively conducts weight estimation and truth inference until a certain convergence criterion is satisfied. The convergence criterion is that the decrease in the objective function is small enough compared with the previous iterations.

Since workers are voluntary in choosing sensing tasks, we cannot guarantee that each task has proxy workers or the UAV for data collection. When the n -th task data is neither collected by the UAV nor by proxy workers, the estimation is obtained by minimizing deviation from the truths to the multi-worker input. The truth discovery process is similar to Algorithm 1. The difference is that we will iteratively conduct weight estimation and truth inference according to the data submitted by all workers recruited in this case.

Algorithm 1 Truth Discovery

Require: Sensory data $\{x_n^m\}_{m \in \mathcal{M}_n^p}$
Ensure: Estimate truth x_n^* for the n -th task
1: Initialize workers' weights $\{h_m\}_{m \in \mathcal{M}_n^p}$
2: **while** Convergence criterion is not satisfied **do**
3: Update x_n^* by Eq. (8)
4: Update workers' weights $\{h_m\}_{m \in \mathcal{M}_n^p}$ by Eq. (7)
5: **end while**
6: **return** x_n^*

B. Data Quality Assessment & Trust Update

Data quality is an essential criterion for measuring the accuracy of the sensing data reported by workers and an important indicator for updating workers' trust. Therefore, we first evaluate the quality of the worker's data. Based on this, a data quality-based trust meter is used to update workers'

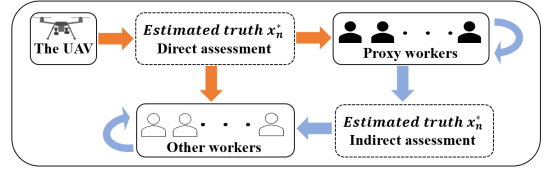


Fig. 2. Data quality assessment.

trust. Workers' trust will increase when workers submit high-quality data. Conversely, when workers submit low-quality data, their trust will decrease.

We consider x_n^* can be used as baseline data to evaluate the quality of data collected by workers, and the closer to the estimation x_n^* , the higher the sensing data quality. Based on the principle, the calculation of data quality is given in Eq. (9). For the convenience of expression, the data quality $q(x_n^m, x_n^*)$ of the data reported by the m -th worker on the n -th task data is represented by $q_{m,n}$.

$$q(x_n^m, x_n^*) = \exp \left(- \left| \frac{x_n^m - x_n^*}{x_n^*} \right| \right). \quad (9)$$

Theorem 1: When x_n^m reported by the m -th worker on the n -th task is closer to x_n^* , the larger $q_{m,n}$ is. On the contrary, when x_n^m reported by the m -th worker on the n -th task is further away from x_n^* , $q_{m,n}$ is smaller, and $0 < q_{m,n} \leq 1$.

Proof: Obviously, when the data x_n^m is closer to x_n^* , that is $|x_n^m - x_n^*| \rightarrow 0$, there is $\lim_{x_n^m \rightarrow x_n^*} q_{m,n} = \lim_{x_n^m \rightarrow x_n^*} \exp \left(- \left| \frac{x_n^m - x_n^*}{x_n^*} \right| \right) = 1$, and when $x_n^m = x_n^*$, $q_{m,n} = 1$; when the worker data x_n^m is farther away from x_n^* , that is $|x_n^m - x_n^*| \rightarrow \infty$, there is $\lim_{|x_n^m - x_n^*| \rightarrow \infty} q_{m,n} = \lim_{|x_n^m - x_n^*| \rightarrow \infty} \exp \left(- \left| \frac{x_n^m - x_n^*}{x_n^*} \right| \right) = 0$. $\square$

As shown in Fig. 2, we use the estimated truth from the data collected by the UAV as baseline data to evaluate the quality of data collected by workers. That is called a direct assessment. The estimated truth from the data collected by proxy workers is used as secondary baseline data to evaluate the data quality. Under the circumstances, an indirect relationship between proxy workers and other workers is established. Direct assessment is accurate but narrow in range. Proxy workers provide a large amount of secondary baseline data for evaluation, but the disadvantage is that it is less accurate than direct evaluation. The data collected by the UAV and proxy workers can be used to evaluate the quality of the data reported by other workers, including proxy workers. At the same time, when the data of the task is neither collected by the UAV nor by proxy workers, we can assess the workers' data quality based on the estimated truth inferred from all of their data. The UAV-assisted split-aggregation truth discovery mechanism can make the estimation more accurate, and the scope of data quality assessment can be expanded.

Data quality and workers' trust complement each other. We can update workers' trust through their data quality. At the same time, after being aware of the workers' trust, we can select trusted workers for a more accurate data quality assessment. Next, we propose a data quality-based trust meter, which updates the trust of workers according to their data quality. In this paper, we consider workers submitting low-quality or

high-quality data as two outcomes of a Bernoulli experiment. Then, the trust of workers can be considered as the result of multiple Bernoulli experiments. The probability of the success of the Bernoulli experiment obeys the Beta distribution, that is, $Beta(\alpha, \beta)$. The probability density function $P(x)$ of the Beta distribution with parameters α and β is:

$$P(x) = \frac{x^{\alpha-1} (1-x)^{\beta-1}}{\int_0^1 y^{\alpha-1} (1-y)^{\beta-1} dy}. \quad (10)$$

The expectation of the Beta distribution is $Ex = \frac{\alpha}{\alpha+\beta}$. Therefore, we use the distribution of Beta (α_m^r, β_m^r) to update the trust of workers, where α_m^r and β_m^r represent parameters of the m -th worker at the r -th trust update, and the trust of the m -th worker after the r -th update is shown in Eq. (11). As no prior knowledge is available, α and β are set to 1 initially, and the initial value of the workers' trust is 0.5.

$$\omega_m^r = \frac{\alpha_m^r}{\alpha_m^r + \beta_m^r}. \quad (11)$$

First, the platform determines the required task data's quality threshold ε . In the data collection process, the data reported by workers are mainly divided into two parts: high-quality and low-quality data. Different trust update strategies are established for two different behaviors: Trust Increase (TI) and Trust Decrease (TD).

Trust Increase (TI): The trust will increase when the quality of data reported by workers is higher than the threshold ε . Meanwhile, considering workers are autonomous in participating in the sensing task, there will be cases where some workers are absent for a long time. The following equation shows the updated hyperparameters α and β when workers report high-quality data and trust decay is considered.

$$\begin{cases} \alpha_m^r = e^{-\Delta t \cdot \varphi} \cdot \alpha_m^{r-1} + \mu \cdot q_{m,n,r}, \\ \beta_m^r = e^{-\Delta t \cdot \varphi} \cdot \beta_m^{r-1}. \end{cases} \quad (12)$$

Among them, $q_{m,n,r}$ represents the data quality of the n -th task reported by the m -th worker at the r -th trust update, and Δt represents the trust update interval. Exponential decay is used in updating α_m^r and β_m^r : $e^{-\Delta t \cdot \varphi}$ on α_m^{r-1} and β_m^{r-1} , where φ is the attenuation factor indicating the decay of trust over time, and μ represents the positive factor. The workers' trust will increase when the quality of the data submitted is above the quality threshold.

Theorem 2: The trust of the m -th worker will increase for any $\mu > 0$ when the m -th worker reports the data quality $q_{m,n,r} \geq \varepsilon$. Moreover, as the number of updates increases, the trust of the m -th worker goes to 1: $\lim_{r \rightarrow \infty} \frac{\alpha_m^r}{\alpha_m^r + \beta_m^r} = 1$.

Proof: First, let's illustrate that when the m -th worker submits high-quality data, the trust of him will increase.

$$\begin{aligned} & \frac{\alpha_m^r}{\alpha_m^r + \beta_m^r} - \frac{\alpha_m^{r-1}}{\alpha_m^{r-1} + \beta_m^{r-1}} \\ &= \frac{\alpha_m^r (\alpha_m^{r-1} + \beta_m^{r-1}) - \alpha_m^{r-1} (\alpha_m^r + \beta_m^r)}{(\alpha_m^r + \beta_m^r) (\alpha_m^{r-1} + \beta_m^{r-1})} \\ &= \frac{\beta_m^{r-1} \cdot \mu \cdot q_{m,n,r}}{(\alpha_m^r + \beta_m^r) (\alpha_m^{r-1} + \beta_m^{r-1})} > 0. \end{aligned}$$

Then, we calculate the limit.

$$\begin{aligned} & \lim_{r \rightarrow \infty} \frac{\alpha_m^r}{\alpha_m^r + \beta_m^r} \\ &= \lim_{r \rightarrow \infty} \frac{e^{-\Delta t \cdot \varphi} \cdot \alpha_m^{r-1} + \mu \cdot q_{m,n,r}}{e^{-\Delta t \cdot \varphi} \cdot \beta_m^{r-1} + e^{-\Delta t \cdot \varphi} \cdot \alpha_m^{r-1} + \mu \cdot q_{m,n,r}} \\ &\leq \lim_{r \rightarrow \infty} \frac{e^{-\varphi \cdot \sum_r \Delta t} \cdot \alpha_m^0 + T_1 \cdot \mu}{e^{-\varphi \cdot \sum_r \Delta t} \cdot (\beta_m^0 + \alpha_m^0) + T_1 \cdot \mu} \\ &\leq \lim_{r \rightarrow \infty} \frac{\alpha_m^0 + T_2}{(\beta_m^0 + \alpha_m^0) + T_2} \\ &\leq \lim_{r \rightarrow \infty} \frac{1 + T_2}{2 + T_2}, \end{aligned}$$

where:

$$\begin{aligned} T_1 &= \left(e^{-\varphi \cdot \sum_{r-1} \Delta t} q_{m,n,1} + \dots + e^{-\varphi \cdot \Delta t} q_{m,n,r-1} + q_{m,n,r} \right), \\ T_2 &= \left(e^{\varphi \cdot \sum_r \Delta t} q_{m,n,r} + \dots + e^{\varphi \cdot \Delta t} q_{m,n,1} \right) \cdot \mu. \end{aligned}$$

When $r \rightarrow \infty$, $\sum_r \Delta t \rightarrow \infty$. we obtain that

$$\lim_{r \rightarrow \infty} \frac{\alpha_m^r}{\alpha_m^r + \beta_m^r} = 1. \quad \square$$

It is not difficult to find from Theorem 2 that when the quality of the data reported by workers is higher than ε , the workers' trust will increase. For the same μ , the higher the quality of the workers' data, the more trust will increase.

Trust Decrease (TD): The trust will decrease when the quality of the data reported by workers is lower than the threshold ε . The following equation shows the updated hyperparameters α and β when workers report low-quality data and trust decay is considered.

$$\begin{cases} \alpha_m^r = e^{-\Delta t \cdot \varphi} \cdot \alpha_m^{r-1}, \\ \beta_m^r = e^{-\Delta t \cdot \varphi} \cdot \beta_m^{r-1} + \delta \cdot (1 - q_{m,n,r}). \end{cases} \quad (13)$$

Among them, $q_{m,n,r}$ represents the quality of the data for the n -th task reported by the m -th worker at the r -th trust update, and δ represents the negative factor. The workers' trust will decrease when the quality of the data submitted is below the quality threshold.

Theorem 3: The trust of the m -th worker will decrease for any $\delta > 0$ when the m -th worker reports the data quality $q_{m,n,r} < \varepsilon$. Moreover, as the number of updates increases, the trust of the m -th worker goes to 0: $\lim_{r \rightarrow \infty} \frac{\alpha_m^r}{\alpha_m^r + \beta_m^r} = 0$.

Proof: First, let's illustrate that when the m -th worker submits low-quality data, the trust of him will decrease.

$$\begin{aligned} & \frac{\alpha_m^r}{\alpha_m^r + \beta_m^r} - \frac{\alpha_m^{r-1}}{\alpha_m^{r-1} + \beta_m^{r-1}} \\ &= \frac{\alpha_m^r (\alpha_m^{r-1} + \beta_m^{r-1}) - \alpha_m^{r-1} (\alpha_m^r + \beta_m^r)}{(\alpha_m^r + \beta_m^r) (\alpha_m^{r-1} + \beta_m^{r-1})} \\ &= \frac{-\alpha_m^{r-1} \cdot \delta \cdot (1 - q_{m,n,r})}{(\alpha_m^r + \beta_m^r) (\alpha_m^{r-1} + \beta_m^{r-1})} < 0. \end{aligned}$$

Then, we calculate the limit.

$$\lim_{r \rightarrow \infty} \frac{\alpha_m^r}{\alpha_m^r + \beta_m^r}$$

$$\begin{aligned}
&= \lim_{r \rightarrow \infty} \frac{e^{-\Delta t \cdot \varphi} \cdot \alpha_m^{r-1}}{e^{-\Delta t \cdot \varphi} \cdot \beta_m^{r-1} + e^{-\Delta t \cdot \varphi} \cdot \alpha_m^{r-1} + \delta \cdot (1 - q_{m,n,r})} \\
&\geq \lim_{r \rightarrow \infty} \frac{e^{-\Delta t \cdot \varphi} \cdot \alpha_m^{r-1}}{e^{-\Delta t \cdot \varphi} \cdot \beta_m^{r-1} + e^{-\Delta t \cdot \varphi} \cdot \alpha_m^{r-1} + \delta} \\
&\geq \lim_{r \rightarrow \infty} \frac{e^{-\varphi \cdot \sum_r \Delta t} \cdot \alpha_m^0}{e^{-\varphi \cdot \sum_r \Delta t} \cdot (\beta_m^0 + \alpha_m^0) + T_3} \\
&\geq \lim_{r \rightarrow \infty} \frac{\alpha_m^0}{(\beta_m^0 + \alpha_m^0) + T_4} \\
&= \lim_{r \rightarrow \infty} \frac{1}{2 + T_4},
\end{aligned}$$

where:

$$\begin{aligned}
T_3 &= \left(e^{-\varphi \cdot \sum_{r-1} \Delta t} + \dots + e^{-\varphi \cdot \Delta t} + 1 \right) \cdot \delta, \\
T_4 &= \left(e^{\varphi \cdot \sum_r \Delta t} + e^{\varphi \cdot \sum_{r-1} \Delta t} + \dots + e^{\varphi \cdot \Delta t} \right) \cdot \delta.
\end{aligned}$$

When $r \rightarrow \infty$, $\sum_r \Delta t \rightarrow \infty$. we obtain that

$$\lim_{r \rightarrow \infty} \frac{\alpha_m^r}{\alpha_m^r + \beta_m^r} = 0.$$

□

It is easy to find from Theorem 3 that when the quality of data reported by workers is lower than the threshold, the workers' trust will decrease. For the same δ , the lower the data quality of the workers' data is, the more the trust value will decrease.

The trust update algorithm based on data quality is shown in Algorithm 2, where $\mathcal{M}_n$ and DBD represent all the workers executing the n -th task and the set of proxy workers, respectively.

Algorithm 2 Data Quality-Based Trust Update

Require: Sensory data $\{x_n^m\}_{m \in \mathcal{M}_n}$

Ensure: Trust value $\mathcal{W} = \{\omega_1, \omega_2, \dots, \omega_M\}$

```

1:  $DBD = \emptyset$ 
2: for  $n$  in  $N$  do
3:   Apply truth discovery algorithm to obtain the estimation  $x_n^*$ .
4:   for  $m$  in  $\mathcal{M}_n$  do
5:     Evaluate worker's data quality through Eq. (9).
6:     Apply trust update mechanism to update trust.
7:     if  $\omega_m^r > \gamma$  and  $m \notin DBD$  then
8:        $DBD = DBD \cup m$ 
9:     end if
10:    if  $\omega_m^r < \gamma$  and  $m \in DBD$  then
11:       $DBD = DBD \setminus m$ 
12:    end if
13:  end for
14: end for
15: return  $\mathcal{W}$ 

```

C. Quality-Aware Trustworthy Incentive Mechanism

Recruiting workers needs to consider two aspects: On the one hand, as data sensing consumes resources such as battery power, computation, and communication resources, a proper incentive mechanism is required to incentive workers to submit high-quality data. On the other hand, we need to reasonably select workers to reward to make full use of the remuneration

budget. Considering the above two aspects, we propose the QTI mechanism. Specifically, when the platform assigns sensing tasks to all idle workers in the sensing range, workers will submit their bids (bid). The bid of the i -th worker is expressed as a tuple $bid_i = (b_i, q_i^*)$, where b_i represents the expected return of the i -th worker, and q_i^* indicates the quality of the data the i -th worker promises to submit. After the platform receives bids from workers, it uses Algorithm 3 to select a set of winning bids and notify them to perform the task. Then, the winning workers perform the sensing task and upload the collected data to the platform. Finally, the platform performs truth discovery after receiving the data reported by the workers and rewards workers based on their data quality.

In designing the selection of the winner set and payment algorithm, we consider that not all workers submit high-quality data truthfully, and there are deviations in the quality of the data they submit. Workers who submit low-quality data can be normal workers submitting low-quality data due to environmental noise or malicious workers who deliberately distort data. We mainly focus on the latter case.

We assume that workers will actively bid to maximize their interests. Based on workers' trust, we design the following algorithm: As shown in Algorithm 3, the inputs include BID and x representing the bidding workers set and the threshold of the number of workers recruited per task, respectively. The output is the winner set S . We first judge whether there are more than x trusted workers in the bidding set BID . If there are less than x , all trusted workers are directly recruited and added to set S ; otherwise, there are x workers with the highest trust level that can be selected from the bidding set BID according to Eq. (14).

Algorithm 3 Selection of the Winner Set

Require: BID, x

Ensure: S

```

1:  $S = \emptyset$ 
2: if The number of trusted workers in  $BID$  is less than  $x$  then
3:   Add all trusted worker to  $S$ 
4: else
5:   while the  $n$ -th task cannot be covered  $x$  times do
6:     select participant  $m$  from  $BID$  according to Eq. (14)
7:     if  $m \notin S$  then
8:        $S = S \cup m$ 
9:        $BID = BID \setminus m$ 
10:    end if
11:  end while
12: end if
13: return  $S$ 

```

Our incentive mechanism uses Eq. (14) to select the worker with the highest trust among all workers in BID as the task performer. Letting ω_i be the i -th workers' trust, we have

$$m = \arg \max_{i \in BID} (\omega_i). \quad (14)$$

After obtaining the winner set, each winner performs the sensing task and reports the results to the platform. The platform obtains the estimation results by the truth discovery

algorithm and then evaluates the quality of each worker according to the Eq. (9). Then, the actual remuneration paid by the platform to the m -th worker is calculated as:

$$p_m = \begin{cases} \min(\widehat{p}_m, b_m) & \text{if } q_m \geq q_m^* \\ \frac{q_m}{q_m^*} \cdot \min(\widehat{p}_m, b_m) & \text{if } q_m^* > q_m > \frac{1}{2} \\ 0 & \text{if } q_m < \frac{1}{2} \end{cases} \quad (15)$$

where $\widehat{p}_m = \max_{j \in S'} \frac{q_m^*}{q_j^*} b_j$ and the j -th worker represents the best choice except the m -th worker in the process of calculating the reward. q_m and q_m^* represent the m -th worker's quality evaluated and the promised data quality, respectively. The specific payment algorithm is shown in Algorithm 4.

Algorithm 4 Payment

Require: BID, S

Ensure: $(p_1, p_2, \dots, p_x)$

```

1: for  $m$  in  $S$  do
2:    $j \leftarrow$  select worker  $m$  from  $BID \setminus m$  by Eq. (14)
3:    $\widehat{p}_m = \max_{j \in S'} \frac{q_m^*}{q_j^*} b_j$ 
4:   Perform truth discovery to get  $x_n^*$ 
5:   Calculate estimated quality  $q_m$  by Eq. (9)
6:   if  $q_m \geq q_m^*$  then
7:      $p_m = \min(\widehat{p}_m, b_m)$ 
8:   else if  $q_m^* > q_m > 1/2$  then
9:      $p_m = \frac{q_m}{q_m^*} \cdot \min(\widehat{p}_m, b_m)$ 
10:  else
11:     $p_m = 0$ 
12:  end if
13: end for
14: return  $(p_1, p_2, \dots, p_x)$ 

```

V. EVALUATION

In this section, we evaluate the performance of the ITD on simulation and real-world datasets. We describe our experimental setup in Section V-A and provide the experimental results in Section V-B.

A. Experiment Setup

We deployed the ITD solution using Python 3.7 on Pycharm and NS3 simulators, conducting simulations on a PC.

1) *Synthetic Dataset:* In our simulation experiments, we built the MCS scene on the Pycharm and NS3 with $M = 200$ workers participating in $N = 100$ sensing tasks. The scenario on the NS3 includes a server, a base station, and 200 mobile workers. The 4G LTE communication network is used for communication between the base station and the mobile worker, and the point-to-point communication network is used between the base station and the server. Throughout the network, we have adopted the IPv4 protocol.

The ground truth x_n^{truth} of each sensing task is sampled from a uniform distribution of $\mathcal{U}(10, 50)$. We simulate normal workers and malicious workers with various information qualities by sampling σ_m^2 from an exponential distribution with parameter λ_1 and with parameter λ_2 , respectively. Therefore, we generate workers' raw sensory data based on the ground

TABLE III
PROPORTION VARIES WITH ε

ε	0.6	0.7	0.8	0.9	0.99
Proportion	0.265	0.176	0.1276	0.0866	0.0204

TABLE IV
PROPORTION VARIES WITH γ

γ	0.6	0.7	0.8	0.9
Proportion (TPA)	0.1313	0.0356	0.0008	0.0
Proportion (PA)	0.0275	0.0023	0.0	0.0

TABLE V
EXPERIMENTAL PARAMETERS

Parameter	Meaning	Value
ε	Quality threshold	0.99
φ	Attenuation factor	0.01
γ	Credit threshold	0.9
λ_1	parameter for normal workers	10
λ_2	parameter for malicious workers	0.01
δ	Negative factor	2
μ	Positive factor	1
Θ	Threshold of tasks collected by UAV	30
b_m	Bid for the task	10 – 25

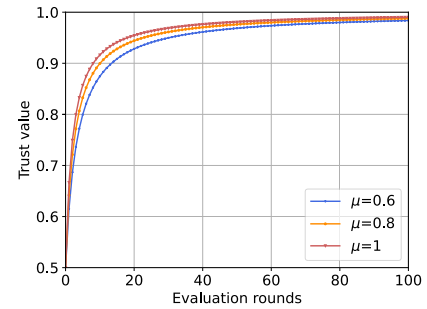


Fig. 3. The trust applying TI.

truths and the sampled error from $\mathcal{N}(0, \sigma_m^2)$. The above simulation settings are the same as those in [22]. In addition, the initial locations of mobile workers are randomly generated, and their location changes follow a random walk model.

Below, we randomly generate 20 scenarios and average the results to ensure the robustness of our solution to different network scenarios. In the experiment parameter, the Quality threshold ε will affect the update of the trust. From TABLE III, it can be seen that as ε increases, the proportion of malicious workers among workers with increased trust will decrease. Moreover, we found that the average data quality submitted by normal workers is greater than 0.99. Therefore, we set $\varepsilon = 0.99$. In addition, we use TPA to represent that malicious workers submitted high-quality data in the first 10% of the tasks in the early stage and carried out probabilistic attacks in the later stage. PA is denoted that malicious workers only launch probabilistic attacks. The result is shown in TABLE IV. It can be found that with the increase of γ , the proportion of malicious workers among proxy workers evaluated will decrease. Therefore, we set $\gamma = 0.9$. Important parameters in our experiment are given in TABLE V.

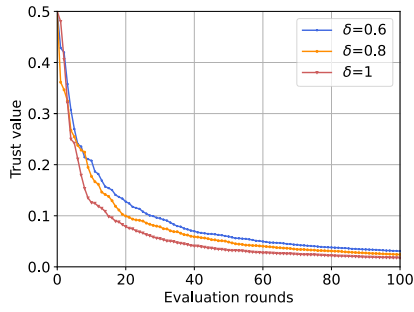


Fig. 4. The trust applying TD.

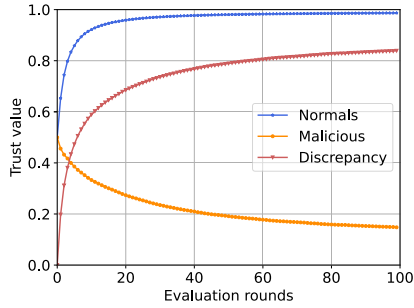


Fig. 5. The trust of workers.

2) *Real-World Dataset*: In addition, we also use two real-world datasets, the noise dataset [48], [49], and the emotion dataset [17] to verify the effectiveness of ITD.

Noise dataset: The noise dataset reveals the state of noise pollution in urban areas. They used mobile clients (mobile phone-based) to collect noise profiles in New York City (NYC). They designed six routes, each containing six measurement locations. For simplicity, we choose the noisy data collected on the six mission points of Route 1 as ground truth. Thus, there are 1614 tasks in total. Worker observations with different degrees of reliability are generated by adding different levels of Gaussian noise on the ground truth with the same settings as the simulated experiments. We plan to have 600 workers to do these tasks.

Emotion dataset: Emotion comprises 38 workers, 700 objects, and 7000 records. Each record is a value from the $[-100, 100]$, signifying the text's emotional intensity (e.g., anger, disgust, and fear). We used some of the data to demonstrate the effectiveness of ITD. We use 663 objects as tasks and execute these tasks in 13 rounds, where each round has 51 tasks. In addition, we set the threshold for the number of UAV acquisition tasks as $\Theta = 15$ in this scene.

3) *Baseline Methods*: To better evaluate the effectiveness of the ITD, we provide the corresponding baseline methods.

We use the Mean Absolute Error (MAE) to illustrate the effectiveness of the ITD in truth discovery. MAE is calculated between the truth discovery result x_n^* and the ground truth x_n^{truth} on N sensing tasks. We compare ITD with the following two baselines.

Conflict Resolution on Heterogeneous (CRH): CRH [20], [21], [22], [38] is a classic truth discovery method in which estimation is iteratively generated from sensory data submitted by all participating workers. In addition, the worker's reward is the lowest price in the bidding.

TABLE VI
PERFORMANCE VARIES WITH THE THRESHOLD Θ

Θ	10	20	30	40	50
Trust of Normals	0.8045	0.8074	0.8078	0.8082	0.8105
Trust of malicious	0.4085	0.4038	0.4036	0.3969	0.3962
MAE	0.1893	0.1654	0.1418	0.1282	0.1208
Cost	3946.86	3972.67	4045.17	4066.70	4089.33

NSITD: To further compare with the ITD, we propose the NSITD to obtain the estimated truth. The difference between the NSITD and the ITD is that the NSITD does not split the workers' reported data according to their trust but obtains the estimated results based on the data of all winners. Moreover, it pays according to the average bid of workers in the winner set.

B. Experimental Results

In this section, we present the experimental results of the ITD and analyze the effectiveness of the ITD from four aspects: trust evaluation, truth discovery, security analysis, and system cost.

1) *Trust Evaluation*: We verify the effectiveness of the trust evaluation mechanism by analyzing the trust between normal and malicious workers. Moreover, the recruitment situation is used to verify the effectuality of the recruitment strategy in the ITD.

Trust update involves two parameters, positive factor μ and negative factor δ . Fig. 3 shows how workers' trust changes under different μ s. It can be seen that when the positive factor μ is larger, the workers' trust grows faster. That also shows that their trust grows faster when workers continue to submit high-quality data. Fig. 4 shows the trend of workers' trust for different δ s. When δ is larger, workers' trust decreases faster. That shows that the more low-quality data submitted, the faster the trust decline rate. This experiment illustrates that our proposed data quality-based trust evaluation scheme is effective.

Distinguishing between normal and malicious workers is the core of trust evaluation. We averaged normal and malicious workers' trust in each round of experiments. As shown in the Fig. 5, the trust of normal workers increases with the evaluation round, in contrast to the decreasing trust trend observed of malicious workers. Experiments show that after 100 rounds of task sensing, the trust of malicious workers is 0.1478, while the trust of normal workers is 0.9868, leading to a trust gap of 0.8390, which kept increasing and reached the maximum after 100 rounds.

The influence of the threshold Θ of the number of tasks collected by the UAV on the evaluation of worker trust is shown in TABLE VI. Among them, we consider the average trust of normal and malicious workers after 100 rounds. We find that the larger Θ is, the larger the average trust of normal workers and the lower the trust of malicious workers. That shows that the larger Θ is, the more obvious the effect of trust evaluation is.

In addition, we also show the growth rate of proxy workers versus the number of tasks and the number of workers.

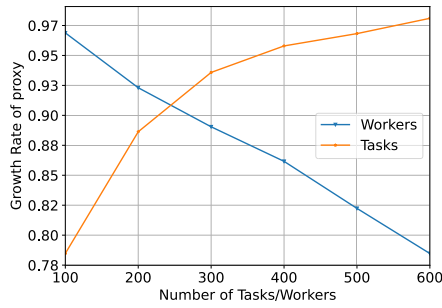


Fig. 6. Growth rate of proxy.

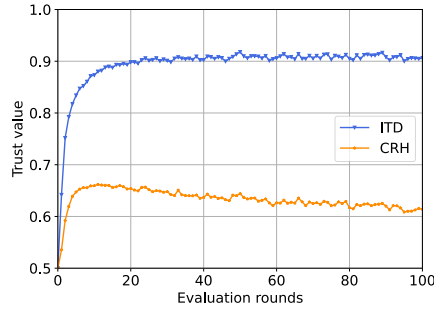


Fig. 7. The trust of participants.

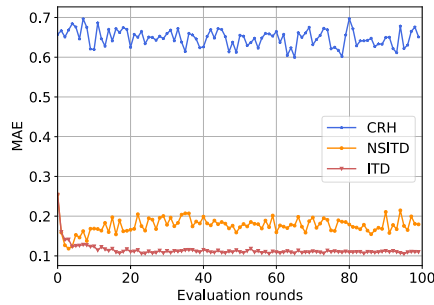


Fig. 8. The simulation datasets.

As shown in Fig. 6, when the number of workers is 200, as the number of tasks increases, the number of tasks completed by the average worker will increase, which makes workers' trust evaluations frequent, resulting in an increase in the growth rate of the proxy. However, when the number of tasks is 300 since the completion of tasks only requires a certain number of workers, this reduces the number of tasks completed by the average worker, reduces the frequency of workers' trust evaluation, and leads to a decrease in the growth rate of proxy.

In recruiting workers to collect sensing data, we hope to recruit credible workers. As shown in Fig. 7, the average trust of recruited workers varies with the evaluation rounds. With the increase in evaluation rounds, the average trust of the recruited workers is also increasing, and the trust of the workers recruited by the ITD has grown faster than the CRH. At the same time, we found that when we adopt the CRH, the average trust of workers is 0.6325, and when we adopt the ITD, the average trust of recruited workers is 0.8920. That shows that recruiting workers based on the trust of workers is effective.

2) *Truth Discovery*: In this part, we discuss and analyze the experimental results of the ITD on truth discovery.

TABLE VII
MAE VARIES WITH THE NUMBER OF TASKS

Task	100	200	300	400	500	600
ITD	0.1245	0.1157	0.1148	0.1145	0.1141	0.1137
NSITD	0.2058	0.1834	0.1882	0.1925	0.1900	0.1908
CRH	0.6850	0.6689	0.6722	0.6680	0.6754	0.6725

TABLE VIII
MAE VARIES WITH THE NUMBER OF WORKERS

Worker	100	200	300	400	500	600
ITD	0.1317	0.1344	0.1393	0.1496	0.1635	0.1668
NSITD	0.2988	0.2517	0.2510	0.2480	0.2336	0.2434
CRH	0.9148	0.8534	0.8166	0.8110	0.8262	0.8086

Here, we discuss the performance of ITD in truth discovery when the proportion of malicious workers among data collectors is 50%. The simulation experiment results about MAE are shown in Fig. 8. The aggregation accuracy of ITD (the smaller the MAE, the higher the aggregation accuracy) is higher than that of CRH and NSITD. ITD is better than CRH, which shows that malicious workers have a great influence on MAE and also fully demonstrates the necessity of identifying malicious workers. The ITD is superior to the NSITD, which shows the necessity of splitting the data based on the trust of winners when the proportion of malicious workers is 50%, which is more significant for improving aggregation accuracy. Moreover, we found that in the early stage of evaluation, the MAE of ITD and NSITD has a downward trend. That is because, in the early stage of evaluation, the platform does not know the trust of workers. However, with the increase of evaluation rounds, the platform has a certain understanding of the trust of workers so that MAE will maintain a stable trend in the later period. The experiment shows a 83.20% increase in the average aggregation accuracy of our proposed ITD compared with the CRH and a 40.39% increase in the average aggregation accuracy compared with the NSITD after 100 rounds. Moreover, the change of the MAE of the ITD with the Θ s is shown in TABLE VI. It can be seen that the larger Θ , the smaller the average MAE, the higher the aggregation accuracy.

The variation of MAE with the number of tasks and the number of workers is shown in TABLE VII and TABLE VIII. The variation of MAE with the number of tasks is shown in TABLE VII when the specified number of workers is 300. We find that as the number of tasks increases, the ITD is always superior to the CRH and the NSITD. Moreover, as the number of tasks increases, the MAE obtained by ITD is smaller. That is because the average evaluation times of workers increase due to the increased number of tasks, making the evaluation more accurate, so the obtained MAE will be smaller. Similarly, the variation of MAE with the number of workers when the number of tasks is specified as 300 is shown in TABLE VIII. We can find that the ITD is always superior to the CRH and the NSITD as the number of workers increases. That shows that in the MCS network, the ITD has better scalability to a certain extent.

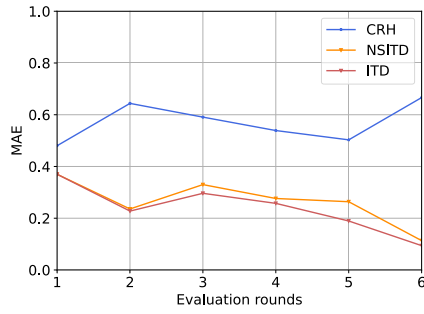


Fig. 9. The noise datasets.

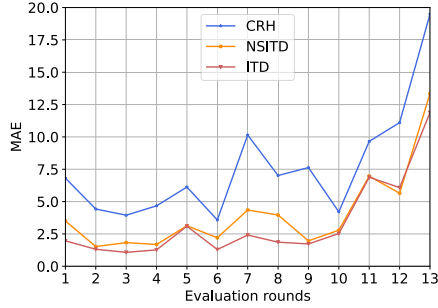


Fig. 10. The emotion datasets.

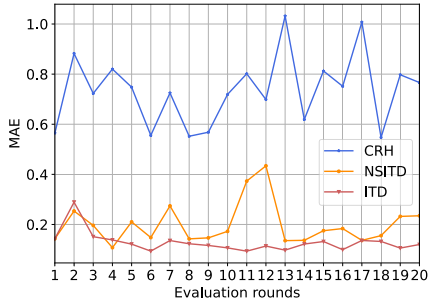


Fig. 11. MAE in NS3.

MAE results for real datasets are shown in Fig. 9 and Fig. 10. Among them, we show the experimental results of MAE in the noisy dataset in Fig. 9. Obviously, the aggregation accuracy of ITD is higher than that of CRH and NSITD. Experiments show that after six rounds, the average aggregation accuracy of ITD is 58.09% higher than that of CRH and 9.7% higher than that of NSITD. Fig. 10 shows the experimental results of MAE on the emotion dataset. Experiments show that after thirteen rounds, the average aggregation accuracy of ITD is 56.11% higher than that of CRH and 18.01% higher than that of NSITD. That shows the superiority of the ITD we proposed. In addition, the result of the NS3 is shown in Fig. 11. The experiment was carried out for 20 rounds, and we found that the MAE obtained by ITD is smaller than that obtained by NSITD and CRH. Specifically, ITD improves the accuracy of truth discovery by 82.51% on average over CRH and 35.61% over NSITD, which shows the effectiveness of our proposed scheme.

3) *Security Analysis of ITD*: In this experiment, there are a total of 200 data collectors to complete 100 tasks. We will consider the proportion of malicious workers that may exist in the data collectors, denoted as 10%, 20%, 30%, 40%, and

TABLE IX
TRUST AND MAE OF ATTACKS

Proportion	Type	Trust	ITD	NSITD	CRH
10%	Probabilistic	0.4082	0.1023	0.0974	0.1175
	On-off attack	0.4998	0.0301	0.0276	0.0531
	Data tampering	0.3472	0.0291	0.0274	0.1639
20%	Probabilistic	0.3690	0.1025	0.0978	0.1852
	On-off attack	0.5005	0.0324	0.0322	0.3962
	Data tampering	0.2479	0.0306	0.0458	1.5849
30%	Probabilistic	0.2562	0.1068	0.1239	0.4414
	On-off attack	0.4942	0.0387	0.0559	0.9161
	Data tampering	0.1627	0.0376	0.1745	3.6826
40%	Probabilistic	0.2337	0.1113	0.1556	0.5819
	On-off attack	0.4882	0.0531	0.1965	1.9023
	Data tampering	0.1127	0.0582	0.8114	7.577
50%	Probabilistic	0.2220	0.1140	0.1764	0.6434
	On-off attack	0.4951	0.0904	0.5507	3.1418
	Data tampering	0.0979	0.1800	2.5934	11.7430



Fig. 12. The cost of the ITD.

50%, respectively, and illustrate the effectiveness and security of ITD in resisting these attacks when they launch probabilistic attacks, On-off attack, and data tampering. As shown in TABLE IX, under different proportions of malicious workers, the highest trust of malicious workers obtained under the probabilistic attack is 0.4082, and the highest trust of malicious workers obtained under the on-off attack is 0.5005. The highest trust of the malicious worker finally obtained under the data tampering is 0.3472. That shows that ITD can defend against these attacks. Furthermore, We found that when the proportion of malicious workers is 30%, 40%, and 50%, the MAE obtained by ITD under different attacks is the smallest than that obtained by NSITD and CRH, and the gap is obvious. Although when the proportion of malicious workers is 10% and 20%, NSITD will be slightly better than ITD, the difference is at most 0.0049, and this error is also within the tolerance range of high-quality data. In addition, we find that ITD achieves the highest 98.47% improvement in the accuracy of truth discovery compared to CRH.

4) *System Costs*: The total cost of our system has two parts. Evaluation cost is the cost of sending the UAV and the cost of recruiting workers determined by the Payment algorithm. The cost of dispatching the UAV is determined by Θ . We assume that the cost of the UAV for a sensing task is five times that of a random sample in the cost range of hiring workers.

Fig. 12 shows the cost of the ITD varies with the evaluation rounds, and Fig. 13 shows the cost of the NSITD varies with the evaluation rounds. The result of the total cost comparison between CRH, ITD, and NSITD is shown in Fig. 14. From the experimental results, ITD and NSITD have higher costs in the first few rounds, which is because the UAV was sent to assist

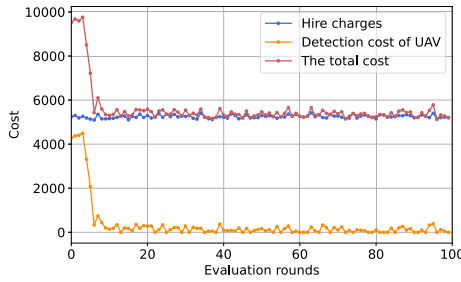


Fig. 13. The cost of the NSITD.

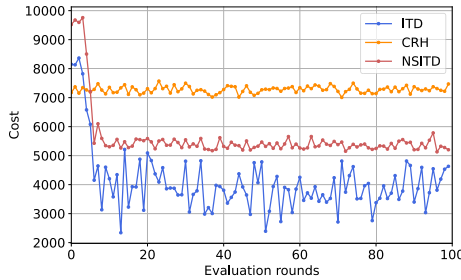


Fig. 14. Comparison of the cost.

in evaluating workers' trust. However, as the evaluation round has increased, many proxy workers have emerged who can assist in evaluating workers' trust. Therefore, the cost remains stable in subsequent rounds. Moreover, we find that the cost is convergent. The variation of system cost with Θ is shown in TABLE VI. We found that the larger Θ is, the higher the cost, but the gap between them is not large. That is because the UAV only collects the data of the task, which that task did not recruit proxy workers. In addition, as shown in Fig. 14, the cost of using ITD is still lower than CRH and NSITD. The cost of the ITD is lower than the cost of the CRH, which shows that it is more cost-effective to use the UAV to assist in evaluating workers' trust to guide the recruitment of a certain number of workers than to recruit enough workers. The cost of ITD is still lower than that of NSITD, which reflects the effectiveness of the payment algorithm of the ITD. Experiments show that compared with CRH, the average cost of ITD is reduced by 43.89%, and compared with the NSITD, the average cost is reduced by 27.14%.

VI. CONCLUSION

With the proliferation of mobile devices with abundant built-in sensors carried by human beings, MCS has been widely used in various fields. The platform needs the data reported by the workers for truth discovery, but malicious workers may report false data, which causes damage. We propose the ITD scheme to select credible workers and incentivize workers to submit high-quality data to the platform. To improve the accuracy of truth discovery, we propose a UAV-assisted split-aggregation truth discovery mechanism to infer the truth. In this process, adding the UAV can improve the estimation accuracy of the truth and assist in evaluating workers' trust. Specifically, we propose to split workers based on their trust and estimate the truth by minimizing deviation from the truth to the multi-proxy worker input. Second,

we evaluate the data quality of the data collected by workers and then propose a data quality-based trust meter to update workers' trust, which lays a foundation for the next round of recruitment. Finally, we propose a quality-aware incentive mechanism to recruit trusted workers and incentive them to submit high-quality data. The experiments show that our proposed ITD reduces costs and improves the accuracy of truth discovery.

In the future, we can consider the following two aspects: On the one hand, workers may add noise when submitting data due to concerns about data privacy disclosure. However, the impact of workers' data privacy on truth discovery and trust evaluation is significant. Future work will focus on discovering accurate truth while protecting worker data privacy. On the other hand, we can explore combining more workers' characteristics, cross-platform historical behavior data, and influencing factors in social networks, such as workers' social relations and social influence to improve accurate predictions of their trust.

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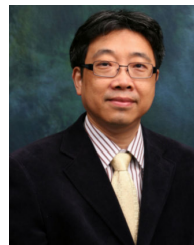
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